

Limit laws, summary statistics, estimation, uncertainty.

Covered by chapter 2.5-2.6 in the textbook.

Laws of large numbers

Let $X_1, X_2, \dots$ be a sequence of independent and identically distributed (iid) random variables where $\mu = E(X)$ and $E|X| < \infty$.

Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Then for every $\epsilon > 0$:

Weak law of large numbers:

$$\lim_{n \rightarrow \infty} P(|\bar{X}_n - \mu| < \epsilon) = 1$$

Strong law of large numbers:

$$P\left(\lim_{n \rightarrow \infty} |\bar{X}_n - \mu| < \epsilon\right) = 1$$

I.e. $\bar{X}_n \rightarrow \mu$ both in probability and almost surely for any distribution.

The important take home message from this is that averages of iid observations always converges towards the expectation!

Implies that the average of many independent simulations of r.v.s converges towards μ .

The central limit theorem

Averages/sums of **many** independent r.v.s (from any distribution!) are approximately normally distributed!

Let $X_1, \dots, X_n$ be independent r.v.s from a distribution with expectation μ and variance σ^2 . We then have (check by yourself!) that $E(\bar{X}) = \mu$ and $\text{Var}(\bar{X}) = \sigma^2/n$.

The central limit theorem (CLT) then says that

$$Z = \frac{\bar{X} - E(\bar{X})}{\sqrt{\text{Var}(\bar{X})}} = \frac{\bar{X} - \mu}{\sqrt{\sigma^2/n}} \approx N(0, 1)$$

Generally OK approximation when $n \geq 30$.

We see that this also implies that

$$\begin{aligned} Z &= \frac{n\bar{X} - n\mu}{n\sqrt{\sigma^2/n}} = \frac{\sum_{i=1}^n X_i - n\mu}{\sqrt{n\sigma^2}} \\ &= \frac{\sum_{i=1}^n X_i - E(\sum_{i=1}^n X_i)}{\sqrt{\text{Var}(\sum_{i=1}^n X_i)}} \approx N(0, 1) \end{aligned}$$

I.e., generally

$$Z = \frac{V - \mathbf{E}(V)}{\sqrt{\text{Var}(V)}} \approx N(0, 1)$$

where $V = \overline{X}$ or $V = \sum_{i=1}^n X_i$.

For an illustration see the R-code: `throw_of_dice.R`

Example: You are throwing a dice 50 times. What is the probability that the sum of all the throws will be larger than 200?

Let X_i be the result of throw number i and $Y = \sum_{i=1}^{50} X_i$. We know that (see discrete uniform distr.) $E(X_i) = 3.5$ and $\text{Var}(X_i) = 2.92$. Since we have a sum of $n = 50$ r.v.s the CLT can be used.

Summary statistics, estimation, uncertainty

Example, robot system:

At an assembly line system three robots are doing different operations on mass produced units. Sometimes the assembly line has to shut down unplanned either due to failures in one of the robots or in the other parts of the assembly line. If we let R_i denote the time until failure in robot i , and A the time to failure in other parts, the time until failure of the entire system is $T = \min\{R_1, R_2, R_3, A\}$.

Assume that the time (in months) until failure for the various parts follow the following distributions:

- $R_1 \sim \text{gamma}(\alpha = 16, \beta = 1)$
- $R_2 \sim \text{gamma}(\alpha = 16, \beta = 1)$
- $R_3 \sim \text{gamma}(\alpha = 12, \beta = 2)$
- $A \sim \text{logN}(\mu = 2.3, \sigma = 1.1)$

Example, robot system (continued):

Given the specifications on the previous page it is easy to simulate many lifetimes $T_1, T_2, \dots, T_n$ from the robot system.

But how should we use these simulations? How should we sum up the results? And how many simulations should we do?

Notice: This is similar to a situation where we gather data on a phenomenon of interest and wonder how much data to gather and how to make useful summaries of the data. The difference is that we generate the data (fast and cheap) on the computer (at the cost of having to know the distribution of the parts of the problem.)

Summary statistics

Some standard summary statistics are:

- Median
- Average (mean of data), $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$
Notice for precision $r = 2\sqrt{\text{Var}(X_i)/n} \Rightarrow n = 4\text{Var}(X_i)/r^2$
- Sample variance, $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$
- Sample standard deviation, $S = \sqrt{S^2}$
- Quantiles

The **upper-tail quantile** with tail probability q is the largest value x_q such that $P(X \geq x_q) \leq q$, i.e.

$$Q^+(q) = \sup\{x_q \in \mathbb{R} | P(X \geq x_q) \leq q\}. \quad (1)$$

Equivalently, at most a fraction q of the distribution lies above $Q^+(q)$.

Remark. An alternative definition, used by R (and the textbook), defines the **lower-tail quantile** as the smallest number x_q such that $P(X \leq x_q) \geq q$, i.e.

$$Q^-(q) = \inf\{x_q \in \mathbb{R} | P(X \leq x_q) \geq q\}. \quad (2)$$

Note: If F is continuous and strictly increasing,

$$Q^-(q) = F^{-1}(q), \quad Q^+(q) = F^{-1}(1 - q),$$

so that

$$Q^+(q) = Q^-(1 - q).$$

By setting the option `lower.tail=FALSE` in R's distribution quantile functions (e.g. `qnorm()`), the upper-tail quantile (1) is returned. This definition is often more convenient in theoretical statistics, for example when deriving confidence intervals.

For data the **empirical quantile** is calculated by first sorting the data such that $x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)}$ and then picking out observation number qn (with some refinements when qn is not an integer.)

For visualization bar plots (for discrete variables) and histograms are very useful.

For continuous variables the **empirical cumulative distribution function (ecdf)** can also be useful. The ecdf is an estimate of the cdf obtained by letting

$$\hat{F}(x) = \begin{cases} 0 & , x < x_{(1)} \\ \frac{i}{n} & , x_{(i)} \leq x < x_{(i+1)} \\ 1 & , x_{(n)} \leq x \end{cases}$$

where $x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)}$ is the ordered sample.

For continuous data it is also possible to find a non-parametric estimate of the pdf by using something called density estimation implemented e.g. in the `density`-function in R.

See the R-example: `robot_system.R`

Estimation

Most of the summary statistics mentioned on the two previous pages can be considered as estimators of (unknown) parameters. E.g. $\bar{X}$ is an estimator of μ , S^2 is an estimator of σ^2 , the empirical quantile is an estimator of the true quantile, and ECDF is an estimator of CDF.

Another estimator we will use a lot is the relative frequency as an estimator of a probability. E.g. to estimate $p = P(X > 7)$ from a sample $X_1, X_2, \dots, X_n$ we use:

$$\hat{p} = \frac{\#X_i > 7}{n}$$

Furthermore, the ecdf is an estimator of the cdf, the bar plot is an estimator of the pmf and the histogram is an estimator of the pdf.

Notice: We use the word **estimator** when we talk about the general procedure (before the experiment) and the word **estimate** when we talk about one specific result (after the experiment).

Definition - Estimator: An estimator is a function of a random sample that is used to estimate an unknown parameter. That is, $\hat{\theta} = g(X_1, \dots, X_n)$, where θ is the parameter of interest. For example, the sample mean $\hat{\mu} = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ is an estimator of the expectation $\mu = E(X)$.

Sometimes it is of interest to estimate other parameters like e.g. the rate λ in the exponential distribution, the shape (α) and scale (β) in the gamma distribution and so on. A general method for finding estimators of unknown parameters in probability distributions is maximum likelihood estimation.

Idea (MLE): Choose the value of θ that makes the observed data $x_1, \dots, x_n$ most likely.

Maximum likelihood estimation (MLE)

Assuming $X_1, \dots, X_n$ i.i.d. $f(x; \theta)$.

1. Define the likelihood function:

$$\begin{aligned} L(\theta; x_1, \dots, x_n) &= f(x_1, \dots, x_n; \theta) \\ &\stackrel{\text{indep.}}{=} f(x_1; \theta) \cdots f(x_n; \theta) \\ &= \prod_{i=1}^n f(x_i; \theta) \end{aligned}$$

The maximum likelihood estimator (MLE) is the value $\hat{\theta} = \arg \max_{\theta} L(\theta; x_1, \dots, x_n)$.

2. Take $\ln()$ (for simpler optimization of $L(\theta)$):

$$l(\theta) = \ln[L(\theta)]$$

3. Take the derivative w.r.t. θ , set equal to 0 and solve:

$$\frac{\partial}{\partial \theta} l(\theta) = 0 \text{ gives } \hat{\theta}$$

4. Check that we have found a maximum:

$$\frac{\partial^2}{\partial \theta^2} l(\theta) < 0 \quad \text{for } \theta = \hat{\theta}$$

Example: Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Exp}(\lambda)$. Find a MLE for λ .

Example: Let $X_1, \dots, X_n$ i.i.d. from a distribution with density

$$f(x, \theta) = \frac{\theta}{x^{\theta+1}}, \quad x > 1.$$

Find a MLE for θ .

Estimate θ given observed data 12.0, 11.2, 13.5, 12.3, 13.8 and 11.9.

Properties of estimators

Definition: An estimator $\hat{\theta}$ is an *unbiased* estimator of a parameter θ if:

$$\text{bias}(\hat{\theta}) = E(\hat{\theta}) - \theta = 0 \quad \square$$

Example: We have seen earlier that $E(\bar{X}) = \mu$.
I.e. $\hat{\mu} = \bar{X}$ is an unbiased estimator of μ .

Definition: Mean square error (MSE):

$$\text{MSE} = E \left[(\hat{\theta} - \theta)^2 \right] = \text{Var}(\hat{\theta}) + \left[\text{bias}(\hat{\theta}) \right]^2 \quad \square$$

A criteria for a good estimator $\hat{\theta}$ is to have small MSE. I.e. no or small bias and as small variance as possible.

Example: We have seen earlier that $\text{Var}(\bar{X}) = \sigma^2/n$. I.e. for $\hat{\mu} = \bar{X}$ we have $\text{MSE}(\hat{\mu}) = \sigma^2/n$.

Uncertainty in estimates/how many simulations should we run?

We can use considerations around MSE for determining how many simulations we need to run. For unbiased estimators this boils down to considerations around variance/standard deviation.

Estimating $\mu = E(X)$:

With $\hat{\mu} = \bar{X}$ we have that $\text{Var}(\bar{X}) = \sigma^2/n$ and thus $\text{SD}(\bar{X}) = \sigma/\sqrt{n}$.

Since by the CLT we have that $\hat{\mu} = \bar{X}$ approximately follows a normal distribution we have (with high precision when n is large) that:

$$P(-z_{\alpha/2} < \frac{\bar{X} - E(\bar{X})}{\text{SD}(\bar{X})} < z_{\alpha/2}) \approx 1 - \alpha$$

$$P(-z_{\alpha/2} < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < z_{\alpha/2}) \approx 1 - \alpha$$

$$P(\mu - z_{\alpha/2}\sigma/\sqrt{n} < \bar{X} < \mu + z_{\alpha/2}\sigma/\sqrt{n}) \approx 1 - \alpha$$

I.e. there is approximately a $(1 - \alpha)100\%$ chance that $\hat{\mu} = \overline{X}$ falls in:

$$\left[\mu \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right]$$

I.e. to have a $(1 - \alpha)100\%$ probability that $\overline{X}$ falls at most a margin e from μ we need:

$$z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < e \quad \Rightarrow \quad n > \frac{z_{\alpha/2}^2 \sigma^2}{e^2}$$

For instance, for a 95% probability, since $z_{0.025} = 1.96 \approx 2$ we need:

$$n > \frac{4\sigma^2}{e^2}$$

But what about σ^2 ? First run a moderate size simulation to estimate σ^2 , and then plug this estimate into the formula.

Example: In the robot system example, how many simulations do we need to run to estimate the expected time to failure with a precision e of respectively, i) one month, ii) one day ?

Generalization - Estimating $E\{h(X)\}$:

Many quantities of interest can be written as

$$\theta = E\{h(X)\}$$

for some function h . Given an iid sample $X_1, \dots, X_n$, a natural estimator is

$$\hat{\theta} = \frac{1}{n} \sum_{i=1}^n h(X_i).$$

By the law of large numbers,

$$\hat{\theta} \longrightarrow E\{h(X)\},$$

and, provided $\text{Var}\{h(X)\} < \infty$, the CLT gives

$$\frac{\hat{\theta} - \theta}{\sqrt{\text{Var}\{h(X)\}/n}} \approx N(0, 1).$$

Thus an estimated standard error is

$$\widehat{\text{SE}}(\hat{\theta}) = \frac{s_h}{\sqrt{n}},$$

where s_h is the sample standard deviation of $h(X_1), \dots, h(X_n)$.

Thus, to have an approximate $(1 - \alpha)100\%$ probability that $\hat{\theta}$ is within a margin e of θ , we need

$$z_{\alpha/2} \frac{\sqrt{\text{Var}\{h(X)\}}}{\sqrt{n}} < e,$$

i.e.

$$n > \frac{z_{\alpha/2}^2 \text{Var}\{h(X)\}}{e^2}.$$

Examples:

$$h(x) = x \quad \implies \quad E\{h(X)\} = E(X),$$

$$h(x) = I(x \in A) \quad \implies \quad E\{h(X)\} = P(X \in A),$$

$$h(x) = \cos x \quad \implies \quad E\{h(X)\} = E(\cos X).$$

Estimating probabilities:

$$P(X \in A) = E\{I(X \in A)\}$$

We estimate the probability p of an event of interest with the relative frequency

$$\hat{p} = \frac{\text{\#successes}}{n} = \frac{1}{n} \sum_{i=1}^n I(X_i \in A)$$

Let $Y = \# \text{successes}$. With independent observations $Y \sim \text{Bin}(n, p)$ (why?).

$$\mathbb{E}(\hat{p}) = \mathbb{E}\left(\frac{Y}{n}\right) = \frac{\mathbb{E}(Y)}{n} = p$$

$$\text{Var}(\hat{p}) = \text{Var}\left(\frac{Y}{n}\right) = \frac{1}{n^2} \text{Var}(Y) = \frac{p(1-p)}{n}$$

Notice that $\hat{p}$ is an average, thus CLT applies and with good precision there is a $(1 - \alpha)100\%$ probability that $\hat{p}$ falls in:

$$[\mathbb{E}(\hat{p}) \pm z_{\alpha/2} \text{SD}(\hat{p})] = \left[p \pm z_{\alpha/2} \frac{\sqrt{p(1-p)}}{\sqrt{n}} \right]$$

I.e. to have a $(1 - \alpha)100\%$ probability that $\hat{p}$ falls at most a margin e from p we need:

$$z_{\alpha/2} \frac{\sqrt{p(1-p)}}{\sqrt{n}} < e \quad \Rightarrow \quad n > \frac{z_{\alpha/2}^2 p(1-p)}{e^2}$$

Further notice that since $p(1 - p) < 0.25$ we will always have a sufficient margin if we use:

$$n > \frac{z_{\alpha/2}^2 \cdot 0.25}{e^2} = \frac{z_{\alpha/2}^2}{4e^2}$$

For instance, for a 95% probability, since $z_{0.025} = 1.96 \approx 2$ we need:

$$n > \frac{1}{e^2}$$

This can be used for all probability simulations. Notice that the ecdf $\hat{F}(x)$ is an estimate of a probability $F(x) = P(X \leq x)$ at any point x , and the precision consideration above thus also holds for each point x of the ecdf. Similarly it also holds point-wise for estimating pmfs.

Example: How many simulations do we need to estimate the probability for the robot system to fail within 1 year with an error of at most:
i) 0.05, ii) 0.005?

Note 1: The standard deviation of an estimator is often called its *standard error* (see the book), $\text{SE}(\hat{\theta}) := \text{SD}(\hat{\theta})$. For example, $\text{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}$ is often called the standard error of the average.

When $\hat{\theta}$ is calculated from Monte Carlo simulations, the corresponding standard error is often called the *Monte Carlo standard error (MCSE)*.

Note 2: In English average refers to $\bar{X}$ and expectation refers to μ , while *mean* can either refer to $\bar{X}$ or μ depending on the context.

Means beyond Euclidean space

For observations $\mathbf{X}_1, \dots, \mathbf{X}_n \in \mathbb{R}^p$, the ordinary sample mean can also be characterized as

$$\bar{\mathbf{X}} = \arg \min_{\mathbf{m} \in \mathbb{R}^p} \frac{1}{n} \sum_{i=1}^n \|\mathbf{X}_i - \mathbf{m}\|^2.$$

This suggests a more general definition.

If the observations lie in a metric space $(\mathcal{M}, d)$, the *sample Fréchet mean* is

$$\hat{\mu}_F = \arg \min_{\mathbf{m} \in \mathcal{M}} \frac{1}{n} \sum_{i=1}^n d^2(\mathbf{X}_i, \mathbf{m}).$$

The corresponding population Fréchet mean is

$$\mu_F = \arg \min_{\mathbf{m} \in \mathcal{M}} E \{ d^2(\mathbf{X}, \mathbf{m}) \}.$$

Thus, the usual sample mean is a special case of the Fréchet mean.

For directional data $\mathbf{X}_i \in S^{p-1}$, one possible choice is the geodesic distance

$$d(\mathbf{x}, \mathbf{m}) = \arccos(\mathbf{x}^T \mathbf{m}).$$

The corresponding Fréchet mean is a mean direction on the sphere.

Using the geodesic distance, the Fréchet mean gives an *intrinsic mean* that respects the geometry of the sphere, whereas, whereas $\frac{\bar{\mathbf{X}}}{\|\bar{\mathbf{X}}\|}$ gives an *extrinsic mean*.

See R-example: `SphereDist.R`